Diabetes Prediction Using an Artificial Neural Network with a Calibrated, Explainable Ensemble

Interim Project Report (Work Completed: Approximately 70%)

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Abstract

Diabetes mellitus is a chronic metabolic condition whose early identification depends on routinely collected measurements such as plasma glucose, body mass index and age. Machine learning can convert these measurements into a risk score, but published studies on the small Pima Indians Diabetes benchmark are difficult to compare because they differ in preprocessing, data partitioning and the way performance is selected and reported. This project builds a transparent and reproducible workflow around an artificial neural network (ANN) and compares it under identical conditions with four established learners and a no-skill baseline.

The dataset contains 768 records with eight numeric predictors and a binary Outcome label (500 negative, 268 positive). The records are split once, with a fixed seed and stratification, into 460 training, 154 validation and 154 test rows. Physiologically impossible zero values in glucose, blood pressure, skin thickness, insulin and BMI are treated as missing. Median imputation with missingness indicators, standardisation and ANOVA feature selection are fitted inside every cross-validation fold so that no information leaks from held-out data. Three feature representations (original, engineered and five selected features) are compared by five-fold cross-validation. Six randomised hyperparameter configurations are searched for each of five models: logistic regression, random forest, histogram gradient boosting, an RBF support vector machine and a multilayer perceptron with one hidden layer of 32 ReLU units, a sigmoid output, the Adam optimiser, L2 regularisation and early stopping.

At this stage the data audit, leakage-controlled pipeline, representation study and model tuning are complete. The tuned models reach cross-validated ROC-AUC between 0.814 and 0.821, with the ANN and logistic regression tied at 0.821. The five-feature representation was selected, engineered features were found to destabilise the ANN,

and the differences between model families are smaller than the fold-to-fold standard deviation. The soft-voting ensemble, probability calibration, validation-set threshold selection, single held-out test evaluation, Shapley explanations and the local web application are the remaining work and are described in the methodology as planned stages. The test set has not been used.

Keywords: diabetes prediction, artificial neural network, Pima Indians dataset, soft-voting ensemble, probability calibration, Shapley explanations.

1. Introduction

1.1 Background and motivation

Diabetes mellitus is a group of metabolic disorders characterised by persistently elevated blood glucose. Left undetected it contributes to cardiovascular disease, kidney failure, neuropathy and retinopathy, and the World Health Organization and the International Diabetes Federation both describe its prevalence as rising worldwide [1], [2]. Because many people remain undiagnosed for years, inexpensive screening aids that use measurements already collected in primary care are of practical interest. A model that ranks individuals by estimated risk could help prioritise confirmatory laboratory testing, although it can never replace a clinical diagnosis.

Supervised machine learning is a natural tool for this task. Artificial neural networks can represent non-linear interactions between predictors, but on small tabular datasets they are often matched or beaten by simpler models. A responsible study therefore has to compare the network with strong baselines under identical conditions rather than reporting it in isolation.

1.2 Problem statement

The task is binary classification of the historical Outcome label in the Pima Indians Diabetes Database from eight numeric inputs. The label is an existing dataset annotation; the project does not forecast future disease onset, and its predictions are not diagnoses. The challenge is to obtain reliable estimates of generalisation on only 768 records, to cope with measurements that are recorded as zero when they are actually missing, and to report both discrimination and the quality of the predicted probabilities.

1.3 Objectives

- To audit the data, quantify missing and invalid values and explore the relationship between each predictor and the outcome using training data only. **(Completed)**

- To design a leakage-controlled pipeline in which imputation, scaling and feature selection are learned only from training folds. **(Completed)**
- To train and tune an ANN and compare it with logistic regression, random forest, gradient boosting, a support vector machine and a no-skill dummy classifier. **(Completed on cross-validation)**
- To combine the models into a soft-voting ensemble, calibrate its probabilities and choose a decision threshold on a separate validation set. **(In progress)**
- To evaluate the frozen model once on an untouched test set with confidence intervals and a paired statistical comparison. **(Planned)**
- To perform ablation studies of representation, model, ensembling, calibration and threshold, and to compare the results with published work. **(Representation ablation completed; remainder planned)**
- To explain predictions with exact Shapley values, permutation importance and what-if analysis, and to deliver a local application. **(Planned)**

1.4 Contributions

The reporting follows the spirit of the TRIPOD guideline for prediction-model studies [35]. The study does not propose a new learning algorithm. Its intended contribution is the careful integration of established techniques into one reproducible protocol, an ablation that isolates the effect of each design decision, and an honest report of the outcome, including cases in which the more elaborate choice did not help. One such negative result is already available: engineered features did not win the cross-validation comparison.

1.5 Organisation of the report

Section 2 surveys related work. Section 3 presents the proposed methodology with a block schematic, equations and algorithms. Section 4 describes the dataset, the experimental setup and the results obtained so far. Section 5 describes the work in progress and the remaining plan. Section 6 gives the interim conclusion, and the references follow.

2. Literature Survey

The Pima Indians Diabetes Database is one of the most frequently used small benchmarks in medical machine learning. It was assembled by the National Institute of Diabetes and Digestive and Kidney Diseases and was used in the late 1980s to evaluate the ADAP neural-style learning algorithm [3]. Since then dozens of studies have applied classical classifiers, ensembles and neural networks to it.

2.1 Diabetes prediction with classical learners

Sisodia and Sisodia compared decision trees, support vector machines and Naive Bayes on the Pima data using 10-fold cross-validation and reported Naive Bayes as best with about 76.3% accuracy [4]. Their work shows that simple probabilistic models are competitive, but it reports a single accuracy figure and does not examine calibration or threshold choice. Zou et al. applied decision trees, random forest and a neural network to hospital physical-examination data and the Pima data, using principal component analysis and minimum redundancy maximum relevance; random forest performed best on their hospital data [5]. Sneha and Gangil found that an optimal feature subset improved several classifiers [6]. Kavakiotis et al. surveyed machine learning in diabetes research and observed that supervised classification, particularly support vector machines, dominates the literature while validation practices vary widely [7].

2.2 Ensembles, boosting and neural networks

Breiman's random forests average many decorrelated decision trees and are robust on small tabular data [8]. Friedman's gradient boosting builds an additive model stage by stage by fitting the negative gradient of a loss, and histogram-based variants make it fast and accurate [9], [10]. Support vector machines maximise the margin of a separating surface and use kernels for non-linearity [11]. Multilayer perceptrons trained by back-propagation [12] with the Adam optimiser [13] and rectified linear units [14] are the standard feed-forward networks, and regularisation by weight decay, dropout and early stopping limits over-fitting on small samples [15]. Grinsztajn et al. showed that tree-based models still frequently outperform deep networks on medium-sized tabular data, which motivates comparing the ANN against forests and boosting rather than assuming its superiority [16].

2.3 Probability calibration and decision thresholds

A classifier that ranks well can still output poorly calibrated probabilities. Platt scaling fits a sigmoid to classifier scores [17], and Niculescu-Mizil and Caruana compared calibration methods across learners [18]. Guo et al. showed that modern neural networks can be over-confident [19]. The Brier score is a proper scoring rule that rewards both discrimination and calibration [20], whereas binned expected calibration error depends on the number of bins, so the two can disagree. Choosing a decision threshold on data separate from the test set is important because the default 0.5 rarely matches the desired balance of sensitivity and specificity, particularly with a 35% positive prevalence [21].

2.4 Explainability in diabetes prediction

Lundberg and Lee unified several attribution methods as SHapley Additive exPlanations [22], and Ribeiro et al. proposed LIME [23]. Tasin et al. combined ensemble classifiers with SHAP and LIME for diabetes prediction [24], and Hasan et al. combined automated machine learning with explainable AI and a Streamlit interface [25]. Permutation importance offers a model-agnostic global view [8], [26]. These studies motivate explaining the complete deployed pipeline, including preprocessing and calibration, rather than an isolated surrogate model.

2.5 Evaluation methodology

Information leakage from preprocessing performed before splitting, repeated tuning on the same folds and selection of the best-looking model on the test set all inflate reported performance. Varma and Simon demonstrated the bias of non-nested cross-validation when it is used for model selection [27], and Cawley and Talbot made the same point for hyperparameter tuning [28]. Efron's bootstrap provides interval estimates [29], and McNemar's test compares paired classifier errors [30]. Saito and Rehmsmeier argued that precision-recall measures are more informative than ROC curves on imbalanced data [31]. This project applies these recommendations.

2.6 Comparison of related work

Ref.	Data	Methods	Validation	Reported result	Limitation addressed here
[3] Smith 1988	Pima	ADAP neural-style learner	Train / test split	About 76% (as commonly cited)	Early work; no calibration or explanation
[4] Sisodia 2018	Pima	DT, SVM, Naive Bayes	10-fold CV	NB accuracy 76.3%	Single metric; no threshold or calibration study
[5] Zou 2018	Luzhou hospital + Pima	RF, DT, NN; PCA, mRMR	Cross-validation	RF best on hospital data	Feature reduction chosen on full data
[6] Sneha 2019	Diabetes records	Feature selection + classifiers	Hold-out / CV	Selection improved accuracy	No probability quality or explanation
[7] Kavakiotis 2017	Review	SVM and others	Various	Qualitative survey	Highlights inconsistent validation
[24] Tasin 2023	Pima + private	Ensembles, SHAP, LIME	Hold-out	High accuracy after resampling	Synthetic resampling; no frozen threshold

Ref.	Data	Methods	Validation	Reported result	Limitation addressed here
[25] Hasan 2025	Pima	AutoML, XAI, Streamlit	Hold-out	AutoML best model	No separate threshold set; no ablation
[16] Grinsztajn 2022	Tabular benchmarks	Trees vs deep nets	Large benchmark	Trees often better	Not diabetes-specific
[19] Guo 2017	Image / text	Temperature scaling	Hold-out	Deep nets over-confident	Not tabular medical data
This work	Pima (768)	ANN + 4 learners + calibrated ensemble	Train / val / test, 5-fold CV, bootstrap	Cross-validated AUC 0.814 to 0.821 (interim)	Reports negative findings and limitations

Earlier studies usually emphasise a single accuracy number, seldom separate the data used for tuning from the data used for threshold selection, and rarely examine calibration or explain the complete pipeline. Reported accuracies cannot be ranked directly against each other because they depend on partitions, preprocessing and sometimes on synthetic resampling.

2.7 Research gap

The reviewed work leaves three gaps that this project addresses. First, the effect of each individual design decision (representation, model family, ensembling, calibration and threshold) is rarely isolated under a fixed protocol. Second, probability quality is seldom reported next to discrimination. Third, explanations are usually produced for a stand-alone classifier rather than for the entire deployed pipeline, so they may not match what the application actually computes.

3. Proposed Methodology

3.1 System overview

The workflow separates three roles for data. The training set (460 rows) is used for exploratory analysis, representation selection, hyperparameter tuning, calibration and final fitting. The validation set (154 rows) is used only to choose the decision threshold. The test set (154 rows) will be used once, after the deployment bundle is frozen.

The pipeline runs in this order: Pima dataset (768 records) → schema validation and stratified split (seed 42) → training, validation and test partitions → fold-local preprocessing (zero handling, imputation, scaling) → feature selection and tuning (5

predictors, 5 learners, 5-fold CV) → ensemble and calibration (equal voting, out-of-fold sigmoid) → threshold selection on validation → frozen bundle → one-shot held-out evaluation and local application. Stages up to and including tuning are complete.

3.2 Data preparation and leakage control

Numeric coercion, binary-label checks and rejection of negative, infinite or duplicate feature records precede splitting, so that duplicates cannot straddle partitions. The split is a seeded (seed 42) stratified 60/20/20 partition. A zero in Glucose, BloodPressure, SkinThickness, Insulin or BMI is physiologically impossible and is replaced by a missing value, while a zero in Pregnancies is valid and retained. Median imputation, standardisation and feature selection are learned on the training portion of each fold. For a feature value x with fold-training median m, mean μ and standard deviation σ, the transformed value is

$z = (\tilde{x} - \mu) / \sigma$, where $\tilde{x} = x$ if observed, otherwise m (1)

A binary missingness indicator is appended for each feature that contains missing values so that the model can learn from the pattern of missingness, which is not assumed to be random. Outliers identified by the 1.5×IQR rule are counted and retained because extreme clinical values are not automatically errors. No synthetic oversampling is used; class weighting is offered as a tuning option for logistic regression, random forest and SVM.

3.3 Feature representations and selection

Three representations are compared. The original representation uses all eight predictors with the indicators. The engineered representation adds log(1+Insulin), log(1+DiabetesPedigreeFunction) and the interaction Glucose×BMI/100. The selected representation keeps the k = 5 features with the largest ANOVA F score computed on training folds. For feature j the F statistic compares between-class and within-class variance:

$F_j = [\Sigma_c n_c (mean_{jc} - mean_j)^2 / (C - 1)] / [\Sigma_c \Sigma_i (x_{ij} - mean_{jc})^2 / (N - C)]$ (2)

where C = 2 is the number of classes and n_c the class size. The representation with the highest mean cross-validated AUC across a fixed logistic and a fixed ANN learner is used for all models.

3.4 Artificial neural network

The ANN is a multilayer perceptron with one hidden layer. For input vector x the hidden activations h and the output probability p are

$h = \mathrm{ReLU}(W_1 x + b_1)$, $\mathrm{ReLU}(u) = \max(0, u)$ (3)

$$p = \sigma(W_2 h + b_2) = 1 / (1 + \exp(-(W_2 h + b_2))) \quad (4)$$

Training minimises the L2-regularised binary cross-entropy over N training examples:

$$L(\theta) = -(1/N) \Sigma_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + (\alpha / 2N) ||\theta||^2 \quad (5)$$

Parameters are updated by Adam [13], which keeps exponential moving averages of the gradient g_t and of its square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \; v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (6)$$

$$\theta_t = \theta_{t-1} - \eta \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon) \quad (7)$$

where $\hat{m}$ and $\hat{v}$ are bias-corrected moments and η is the learning rate. Early stopping holds out an internal validation fraction of the training data and stops when its score stops improving. The search selected 32 hidden units, $\eta = 0.003$ and $\alpha = 0.001$.

3.5 Comparison models

Logistic regression models the log-odds linearly and is fitted with L2 regularisation of strength 1/C. The random forest averages bootstrap-trained trees with limited depth and leaf size. Histogram gradient boosting builds an additive model $F_m(x) = F_{m-1}(x) + \nu h_m(x)$, where each tree h_m fits the negative gradient of the log-loss and ν is the learning rate [9], [10]. The support vector machine uses the radial basis kernel

$$K(x, x') = \exp(-\gamma ||x - x'||^2) \quad (8)$$

with probabilities obtained through internal calibration. A dummy classifier predicting the training prevalence exposes the performance achievable without information from the features.

3.6 Hyperparameter search

For each model, six random configurations are evaluated by stratified five-fold cross-validation on the training set with ROC-AUC as the criterion. The ANN is fixed in advance as the comparator for the later statistical test. Because the same folds are used for selecting representation and parameters, the best cross-validation scores are optimistically biased; this is acknowledged, and the untouched test set will provide the independent estimate.